early twentieth century led directly to the silicon transistors buried in your phone and the LEDs in its screen, the nuclear hearts of the most distant space probes and the lasers in the supermarket checkout scanner. Quantum physics explains why the Sun shines and how your eyes can see. It explains the entire discipline of chemistry, periodic table and all. It even explains how things stay solid, like the chair you're sitting in or your own bones and skin. All of this comes down to very tiny objects behaving in very odd ways.

But there's something troubling here. Quantum physics doesn't seem to apply to humans, or to anything at human scale. Our world is a world of people and keys and other ordinary things that can travel down only one path at a time. Yet all the mundane things in the world around us are made of atoms—including you, me, and Danish princes. And those atoms certainly are governed by quantum physics. So how can the physics of atoms differ so wildly from the physics of our world made of atoms? Why is quantum physics only the physics of the ultratiny?

The problem isn't that quantum physics is weird. The world is a wild and wooly place, with plenty of room for weirdness. But we definitely don't see all the strange effects of quantum physics in our daily lives. Why not? Maybe quantum physics really is only the physics of tiny things, and it doesn't apply to large objects—perhaps there's a boundary somewhere, a border beyond which quantum physics doesn't work. In that case, where is the boundary, and how does it work? And if there is no such boundary—if quantum physics really applies to us just as much as it applies to atoms and subatomic particles—then why does quantum physics so flagrantly contradict our experience of the world? Why aren't our keys ever in two places at once?

Eighty years ago, one of the founders of quantum physics, Erwin Schrödinger, was deeply troubled by these problems. To explain his concerns to his colleagues, he devised a now-famous thought experiment: Schrödinger's cat (Figure I.1). Schrödinger imagined putting a cat in a box along with a sealed glass vial of cyanide, with a small hammer hanging